Arab Academy for Science and Technology Maritime Transport

College of Computing and Information Technology

Computer Science

Computer Graphics

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1. Introduction

This project demonstrates the implementation of core 2D computer graphics algorithms from scratch using C++ and OpenGL (FreeGLUT).

Instead of relying on built-in high-level graphics library functions, all rasterization, mathematical transformations, and clipping operations were implemented manually to provide a deep understanding of the mathematical and geometric foundations of computer graphics.

The project also includes a modern interactive graphical user interface (GUI), real-time performance metrics, an Undo/Redo system, and scene serialization for saving and loading work.

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2. Objectives

- Implement and analyze line drawing algorithms:
  - DDA (Digital Differential Analyzer)
  - Bresenham Line Algorithm
- Implement circle drawing algorithms:
  - Midpoint Circle Algorithm
  - Bresenham Circle Algorithm
- Apply matrix-based 2D transformations:
  - Translation
  - Rotation
  - Scaling
  - Reflection
  - Shearing
- Implement clipping algorithms:
  - Cohen-Sutherland Line Clipping
  - Sutherland-Hodgman Polygon Clipping
- Generate parametric curves:
  - Cubic Bezier Curves

    - B-Spline Curves
- Compare performance, efficiency, and accuracy of all implemented algorithms.

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3. Algorithm Descriptions

3.1 Line Drawing Algorithms

DDA (Digital Differential Analyzer)

The DDA algorithm uses floating-point increments based on the line equation:

$y = mx + c$

It calculates step values for x and y and rounds each computed coordinate to the nearest pixel.

Advantages:

- Simple to understand and implement.

Disadvantages:

- Slower due to floating-point calculations and rounding operations.

Bresenham Line Algorithm

Bresenham uses an integer decision parameter to determine the next pixel position.

Advantages:

- Avoids floating-point operations.
- Faster and more efficient for raster displays.
- Uses only integer addition and subtraction.

3.2 Circle Drawing Algorithms

Midpoint Circle Algorithm

This algorithm uses the decision function:

$f(x, y) = x^2 + y^2 - r^2$

It determines whether the next pixel should move East or South-East.

Features:

- Calculates only one octant of the circle.
- Uses 8-way symmetry to mirror points and complete the full circle.

Advantages:

- Fast and highly efficient.

Bresenham Circle Algorithm

This algorithm minimizes the error between the selected pixel and the actual circle boundary.

Advantages:

- Stable integer calculations.
- High drawing accuracy.
- Efficient execution.

3.3 2D Transformations

All transformations are implemented mathematically relative to a pivot point.

Translation

Moves an object by adding displacement values:

(dx, dy)

to all vertices.

Rotation

Uses sine and cosine functions to rotate shapes around a pivot point.

Scaling

Resizes objects uniformly or non-uniformly.

Shearing

Slants shapes horizontally or vertically.

Reflection

Reflects objects across:

- X-axis
- Y-axis
- Origin

3.4 Curves and Clipping

Bezier Curves

Cubic Bezier curves are generated using Bernstein polynomials.

Features:

- Smooth curve generation.
- Global control over curve shape.

B-Spline Curves

B-Spline curves use basis functions.

Advantages:

- Local control of shape.
- Modifying one control point does not affect the entire curve.

Disadvantages:

- More computationally intensive.

Cohen-Sutherland Line Clipping

Uses 4-bit outcodes for each point.

Advantages:

- Fast accept/reject operations using bitwise logic.

Sutherland-Hodgman Polygon Clipping

Clips polygons against all four clipping window boundaries sequentially.

Supports:

- Convex polygons
- Complex polygon clipping operations

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4. Performance Analysis

Algorithm	Performance	Notes
DDA Line	Slower	Floating-point operations and rounding overhead
Bresenham Line	Very Fast	Integer-only calculations
Midpoint Circle	Fast	Uses 8-way symmetry
Bezier Curve	Moderate	Iterative calculations over parameter t
B-Spline Curve	Heavy	Basis function computations
Cohen-Sutherland	Fast	Bitwise logical operations

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5. Challenges Faced

- Managing OpenGL and FreeGLUT dependencies.
- Integrating CMake correctly across different systems.
- Synchronizing screen coordinates with OpenGL viewport coordinates.
- Handling floating-point precision in curve generation.
- Implementing polygon clipping edge cases correctly.
- Building Undo/Redo state management.
- Implementing Scene Serialization for Save/Load functionality.

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6. Screenshots

Insert project screenshots in the following sections:

1. Main Interface
2. Transformations Demonstration
3. Clipping Demonstration
4. Split-Screen Comparison

(Replace this section with actual screenshots before submission.)

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7. Conclusion

This project successfully implements all required computer graphics algorithms while providing an interactive environment for experimentation and visualization.

The comparison mode demonstrates the practical efficiency advantages of integer-based algorithms such as Bresenham over floating-point algorithms like DDA.

Additional features such as:

- B-Spline Curves
- Undo/Redo System
- Scene Serialization

extend the project beyond the minimum requirements and transform it into a small graphics engine suitable for educational and experimental purposes.

GitHub: https://github.com/RemonAyman/computergraphics

Presentation link: https://canva.link/gcf2eabkub6o2qp